Every Wave Travels In One Of Two Ways

SYNOPSIS

In Lab 1 you built an oscillation and it went nowhere. The mass bobbed up and down at one spot on the table, the pattern never spread out into space, and that is why there was no wavelength to measure — you had the ingredient of a wave, but not a wave. Now we give that oscillation something it did not have: neighbors. Hand the wiggle to the chunk of stuff next door, let that chunk hand it on again, and the disturbance walks away from you. That is a **moving wave**, and here is the surprise: there are only two ways it can walk. Either the stuff moves back and forth *along* the same line the disturbance travels — we call that **longitudinal** — or the stuff moves *across* that line, which we call **transverse**. You have already met both without knowing it. Pluck a guitar string and the blur you see is the string snapping side to side while the wave runs along its length: transverse. But the sound that leaves that string and arrives at your ear is air being squeezed and stretched in the very direction it is heading: longitudinal. One instrument, one note, two different kinds of wave back to back. Remember the stadium wave from Note 1 — nobody leaves their seat, only the pattern travels. That is still true today. The only new question is which way the spectators stand up: along the direction the wave is sweeping, or across it. A slinky will do either one on demand, which is why there is one on your table.

You will need: A slinky, a partner, a twist-tie or a short piece of ribbon, a stopwatch (your phone works), and a long clear stretch of floor or table.

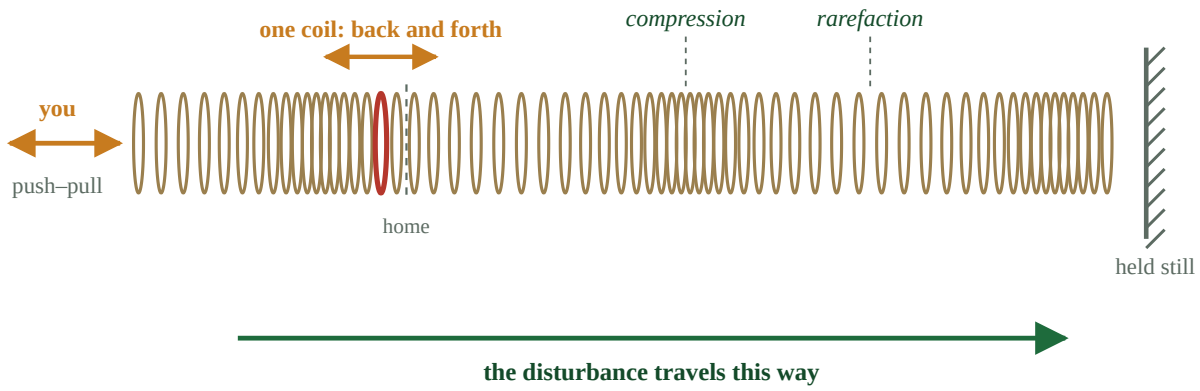


Fig 1. A **longitudinal** wave. You push and pull the near end along the slinky's own length. Crowded coils (a *compression*) and spread-out coils (a *rarefaction*) march toward the far end. Watch the red coil: it slides back and forth about its home spot and stays there. The short double-headed arrow (amber) marks how the *slinky* moves; the long arrow (green) marks how the *disturbance* moves. Here the two are along the same line.

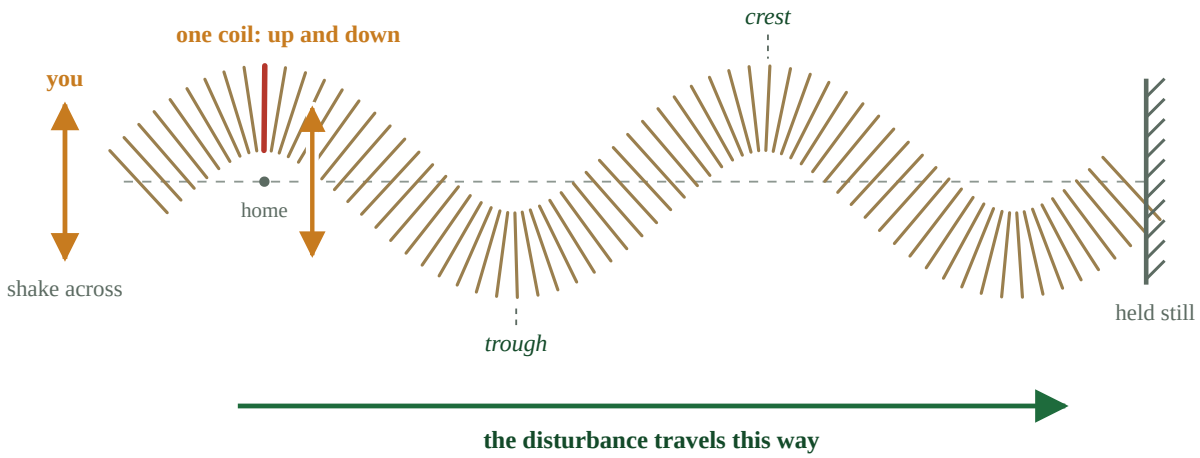


Fig 2. A **transverse** wave on the same slinky. Now you shake the near end across the slinky's length, and *crests* and *troughs* march toward the far end instead. The red coil rides up and down about its home spot and, again, stays there. Same two arrows — but this time they are at right angles.

Performance Expectations

Produce both a longitudinal and a transverse wave on a slinky. Name each one by comparing the direction the medium oscillates with the direction the disturbance travels — not by how the wave looks. Classify sound in air, and show that the speed along the slinky is set by the slinky, not by how hard or how fast you shake it.

Preamble

Before answering anything, get a slinky and a partner and do these five things. Jot down what you notice as you go — the questions later will ask for exactly these observations.

1. Stretch the slinky along the floor or a long table until it is two or three metres long. Your partner holds one end *still*; that end is a wall for the rest of today. Let it settle.

2. Tie a bright twist-tie or a short piece of ribbon onto a single coil near the middle. This is the coil you will spy on. Everything that matters today is visible on that one ribbon.

3. Push your end sharply toward your partner and pull it straight back, once. Watch what leaves your hand and where it goes. Then do it over and over at a steady rhythm and watch the ribbon, not the wave.

4. Now shake your end across the slinky instead — up and down, or side to side on a table. Again: one sharp flick first, then a steady rhythm. Watch the ribbon again.

5. Send one single pulse and have your partner time how long it takes to reach the far end. Repeat it with a much bigger flick, and again with a much faster rhythm. Does the pulse ever arrive sooner?

KEEP THIS PICTURE IN MIND

In both experiments the ribbon ends up back where it started and the far end of the slinky never comes any closer to you. Nothing made the trip except the disturbance. So the question that separates the two names is never "what does the wave look like?" — it is always "which way does the ribbon move, compared with which way the pattern goes?"

Instructions

Look at Fig. 1 and Fig. 2 and reread the Synopsis. Answer in your own words; there is no math here, only careful looking and thinking. Write your answers in the space provided.

1. Start with the push–pull experiment. In a few words, describe how the ribbon moved. Now describe which way the bunched-up region traveled. Are those two directions the same line, or at right angles?

2. Now the across-the-slinky experiment. Describe how the ribbon moved, and which way the crests traveled. Same line, or right angles?

3 Using your two answers, write your own definition of *longitudinal* and *transverse*. (The Latin helps: *longitudo* is length, and *trans-* means across.)

4 In Lab 1 the mass bobbed on its spring and the pattern went nowhere — speed zero, no wavelength to measure. The slinky coils also stay home, yet something clearly travels. What does the slinky have that the single mass on a spring did not?

5 Fig. 1 labels a crowded region a *compression* and a spread-out region a *rarefaction*; Fig. 2 labels the high and low spots a *crest* and a *trough*. Both pairs are just "the disturbance, at its most extreme." Why do we bother giving them different names?

6 Turn to your timing. When you flicked harder, did the pulse reach your partner any sooner? When you shook faster? If the answer is no both times, then what *did* change about the wave — and what does that tell you about what sets the speed?

7 Air can be squeezed, but you cannot grab a fistful of it sideways and drag it the way you can grab a slinky coil. Given that, which of the two kinds of wave can travel through open air? Is sound longitudinal or transverse? Defend your answer by saying what the air does.

8 Go back to the guitar. The string itself carries one kind of wave; the sound that reaches your ear is the other kind. Name each one, and describe what happens at the moment the string hands its energy over to the air.

9 Try to make a wave that is neither — twist the end of the slinky, or shake it in a circle, and see what runs down the coils. Does everything you can produce fall neatly into one of the two names?

10 Put it together. In Note 1 we said a wave is a repeating disturbance that travels. Write that sentence again, but now say more about the disturbance, using the words *medium*, *oscillates* and *travels* — and make your sentence cover both figures in this note.

Draw it yourself

Now draw the two waves from memory — no peeking at Fig. 1 and Fig. 2 until you have tried. In each box, sketch the coils, circle one coil and draw an arrow showing how that coil moves, then draw a separate arrow showing which way the disturbance travels. Label both arrows.

A — Longitudinal

Mark a compression and a rarefaction. Your two arrows should end up along the same line.

B — Transverse

Mark a crest and a trough. Your two arrows should end up at right angles.

WHERE THIS IS GOING

Look at Fig. 1 and Fig. 2 once more and notice something Lab 1 could not show you: the pattern now repeats across *space*. There is a real distance from one compression to the next, and from one crest to the next. That distance is the wavelength that was missing from the oscillator, and the moment we can measure it, $v = \lambda \times f$ stops being a formula on a page and starts being a thing you can check with a stopwatch. That is the next note.

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